CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CE/IT) Examination

November 2012

CE204 Digital Electronics

Date: 29.11.2012, Thursday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

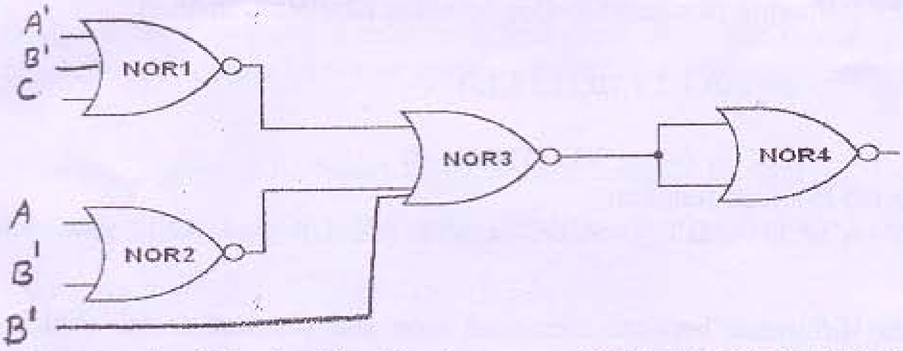
SECTION – I

Q - 1 **(a)** Evaluate $(734)_8 = ()_{16}$ 01

(b) Explain RS latch using NAND gate. 04

(c) Obtain truth table of the function F=xy+xy'+y'z 02

Q - 2 **(a)** Determine the Boolean function for the output F of the circuit given below. 04



(b) Express the following function in a sum of Minterm and product of Maxterm. 04
F(A,B,C,D)=D(A'+B)+B'D

(c) Design a counter with the sequence 0,1,2,4,5,6 and repeat. Use JK flip-flop. 06

OR

Q - 2 **(a)** Explain operation of shift register. 04

(b) Determine the prime implicants of the function 04
F (w, x, y, z) = Σ m (1,4,6,7,8,9,10,11,15)

(c) Reduce the following Boolean expressions to the required number of literals. 06

F=ABC+A'B'C+A'BC+ABC'+A'B'C' (To 5 literals)
F=BC+AC'+AB+BCD (To 4 literals)

Q - 3 **(a)** Implement the following equation with NOR gate using NOR as an universal gate logic. 04
F= A(B+CD)+BC'

(b) Write a short note on Magnitude comparator. 04

(c) Reduce the number of states in the following state table and tabulate the reduced state table along with state assignment. 06

Present state	Next state		Output	
	X=0	X=1	X=0	X=1
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	g	f	0	1
g	a	f	0	1

OR

Q - 3 (a) Design Full Adder with two half adder and an OR gate. 04

(b) Explain BCD ripple counter. 06

(c) Simplify the following Boolean function by using tabulation method. 04

$$F = \Sigma(0,1,2,8,10,11,14,15)$$

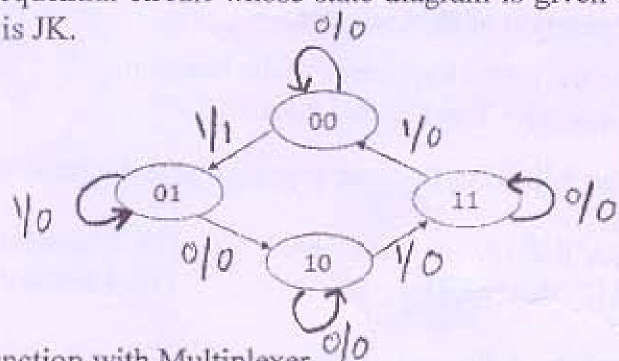
SECTION – II

Q - 4 (a) Implement the Boolean function 03
F=AB'CD'+A'BCD'+AB'C'D+A'BC'D with EX OR and AND gate with logic diagram.

(b) What is the difference between canonical form and standard form? Which form is preferable when implementing a boolean function with gates? Which form is obtained from a truth table? 03

(c) How many don't care inputs are there in BCD Adder. 01

Q - 5 (a) Design a clocked sequential circuit whose state diagram is given below. The type of flip-flop to be used is JK. 07



(b) Implement given function with Multiplexer. 03
F(A,B,C,D)= Σ (0,1,3,4,8,9,15)

(c) (i)Using 10's complement perform M-N with the given decimal numbers 04

M=3250,N=72532

(ii) Using 2's complement perform M-N with the given decimal numbers
M=1000100,N=1010100

OR

Q - 5 (a) A combinational circuit is defined by the functions: 06

F1(A,B,C)= Σ(3,5,6,7)

F2(A,B,C)= Σ(0,2,4,7)

Implement the circuit with a PLA having three inputs, four product terms and two outputs.

(b) Explain the operation of D flipflop along with its symbol, characteristic table and characteristic equation. 04

(c) (i)The content of 4 bit shift register is initially 1101.The register is shifted 4 times to right; with the serial input 101101.What is the content of the register after each shift? 04

(ii) Give the difference between combinational circuit and sequential circuit.

Q - 6 (a) Describe the operation perform by the following logic circuit. 04

(a)Decoder(b)Encoder

(b) Write a short note on Johnson counter. 04

(c) Design a combinational circuit with four input lines that represent decimal digit in BCD and four output lines that generate the 9's complement of input digit. 06

OR

Q - 6 (a) Draw and Explain 4-bit binary ripple counter with JK flipflop. 06

(b) Built the following function in SOP and POS form using kmap. 04

F (w,x,y,z) = $\sum(1,3,7,11,15)$ + d(0,2,5)

(c) Design serial adder using sequential logic procedure. 04

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